scalar X, or a vector or matrix $\mathbf{X}$. In addition, we use $X_{i,j}$ to denote the i, jth element. Thus, $\mathbf{X}$ and `X` (in a MATLAB code fragment) may both refer to the same variable.

Quiz 2.5

The flip of a thick coin yields heads with probability 0.4, tails with probability 0.5, or lands on its edge with probability 0.1. Simulate 100 thick coin flips. Your output should be a 3×1 vector $\mathbf{X}$ such that X_1, X_2, and X_3 are the number of occurrences of heads, tails, and edge.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

2.1.1● Suppose you flip a coin twice. On any flip, the coin comes up heads with probability 1/4. Use H_i and T_i to denote the result of flip i.

(a) What is the probability, $P[H_1|H_2]$, that the first flip is heads given that the second flip is heads?

(b) What is the probability that the first flip is heads and the second flip is tails?

2.1.2● For Example 2.2, suppose $P[G_1] = 1/2$, $P[G_2|G_1] = 3/4$, and $P[G_2|R_1] = 1/4$. Find $P[G_2]$, $P[G_2|G_1]$, and $P[G_1|G_2]$.

2.1.3● At the end of regulation time, a basketball team is trailing by one point and a player goes to the line for two free throws. If the player makes exactly one free throw, the game goes into overtime. The probability that the first free throw is good is 1/2. However, if the first attempt is good, the player relaxes and the second attempt is good with probability 3/4. However, if the player misses the first attempt, the added pressure reduces the success probability to 1/4. What is the probability that the game goes into overtime?

2.1.4● You have two biased coins. Coin A comes up heads with probability 1/4. Coin B comes up heads with probability 3/4. However, you are not sure which is which, so you choose a coin randomly and you flip it. If the flip is heads, you guess that the flipped coin is B; otherwise, you guess that the flipped coin is A. What is the probability $P[C]$ that your guess is correct?

2.1.5■ Suppose that for the general population, 1 in 5000 people carries the human immunodeficiency virus (HIV). A test for the presence of HIV yields either a positive (+) or negative (−) response. Suppose the test gives the correct answer 99% of the time. What is $P[-|H]$, the conditional probability that a person tests negative given that the person does have the HIV virus? What is $P[H|+]$, the conditional probability that a randomly chosen person has the HIV virus given that the person tests positive?

2.1.6■ A machine produces photo detectors in pairs. Tests show that the first photo detector is acceptable with probability 3/5. When the first photo detector is acceptable, the second photo detector is acceptable with probability 4/5. If the first photo detector is defective, the second photo detector is acceptable with probability 2/5.

(a) Find the probability that exactly one photo detector of a pair is acceptable.

(b) Find the probability that both photo detectors in a pair are defective.

2.1.7■ You have two biased coins. Coin A comes up heads with probability 1/4. Coin B comes up heads with probability 3/4. However, you are not sure which is which so you flip each coin once, choosing the first coin randomly. Use H_i and T_i to denote the result of flip i. Let A_1 be the event that coin A was flipped first. Let B_1 be the event that coin B was flipped first. What is $P[H_1H_2]$?

Are H_1 and H_2 independent? Explain your answer.

2.1.8 A particular birth defect of the heart is rare; a newborn infant will have the defect D with probability $P[D] = 10^{-4}$. In the general exam of a newborn, a particular heart arrhythmia A occurs with probability 0.99 in infants with the defect. However, the arrhythmia also appears with probability 0.1 in infants without the defect. When the arrhythmia is present, a lab test for the defect is performed. The result of the lab test is either positive (event T^+) or negative (event T^-). In a newborn with the defect, the lab test is positive with probability $p = 0.999$ independent from test to test. In a newborn without the defect, the lab test is negative with probability $p = 0.999$. If the arrhythmia is present and the test is positive, then heart surgery (event H) is performed.

(a) Given the arrythmia A is present, what is the probability the infant has the defect D?

(b) Given that an infant has the defect, what is the probability $P[H|D]$ that heart surgery is performed?

(c) Given that the infant does not have the defect, what is the probability $q = P[H|D^c]$ that an unnecessary heart surgery is performed?

(d) Find the probability $P[H]$ that an infant has heart surgery performed for the arrythmia.

(e) Given that heart surgery is performed, what is the probability that the newborn does *not* have the defect?

2.1.9 Suppose Dagwood (Blondie's husband) wants to eat a sandwich but needs to go on a diet. Dagwood decides to let the flip of a coin determine whether he eats. Using an unbiased coin, Dagwood will postpone the diet (and go directly to the refrigerator) if either (a) he flips heads on his first flip or (b) he flips tails on the first flip but then proceeds to get two heads out of the next three flips. Note that the first flip is *not* counted in the attempt to win two of three and that Dagwood never performs any unnecessary flips. Let H_i be the event that Dagwood flips heads on try i. Let T_i be the event that tails occurs on flip i.

(a) Draw the tree for this experiment. Label the probabilities of all outcomes.

(b) What are $P[H_3]$ and $P[T_3]$?

(c) Let D be the event that Dagwood must diet. What is $P[D]$? What is $P[H_1|D]$?

(d) Are H_3 and H_2 independent events?

2.1.10 The quality of each pair of photo detectors produced by the machine in Problem 2.1.6 is independent of the quality of every other pair of detectors.

(a) What is the probability of finding no good detectors in a collection of n pairs produced by the machine?

(b) How many pairs of detectors must the machine produce to reach a probability of 0.99 that there will be at least one acceptable photo detector?

2.1.11 In Steven Strogatz's New York Times blog `http://opinionator.blogs.nytimes.com/2010/04/25/chances-are/?ref=opinion`, the following problem was posed to highlight the confusing character of conditional probabilities.

Before going on vacation for a week, you ask your spacey friend to water your ailing plant. Without water, the plant has a 90 percent chance of dying. Even with proper watering, it has a 20 percent chance of dying. And the probability that your friend will forget to water it is 30 percent. (a) What's the chance that your plant will survive the week? (b) If it's dead when you return, what's the chance that your friend forgot to water it? (c) If your friend forgot to water it, what's the chance it'll be dead when you return?

Solve parts (a), (b) and (c) of this problem.

2.1.12 Each time a fisherman casts his line, a fish is caught with probability p, independent of whether a fish is caught on any other cast of the line. The fisherman will fish all day until a fish is caught and then he will quit and go home. Let C_i denote the event that on cast i the fisherman catches a fish. Draw the tree for this experiment and find $P[C_1]$, $P[C_2]$, and $P[C_n]$ as functions of p.

2.2.1● On each turn of the knob, a gumball machine is equally likely to dispense a red, yellow, green or blue gumball, independent from turn to turn. After eight turns, what is the probability $P[R_2Y_2G_2B_2]$ that you have received 2 red, 2 yellow, 2 green and 2 blue gumballs?

2.2.2● A Starburst candy package contains 12 individual candy pieces. Each piece is equally likely to be berry, orange, lemon, or cherry, independent of all other pieces.

(a) What is the probability that a Starburst package has only berry or cherry pieces and zero orange or lemon pieces?

(b) What is the probability that a Starburst package has no cherry pieces?

(c) What is the probability $P[F_1]$ that all twelve pieces of your Starburst are the same flavor?

2.2.3● Your Starburst candy has 12 pieces, three pieces of each of four flavors: berry, lemon, orange, and cherry, arranged in a random order in the pack. You draw the first three pieces from the pack.

(a) What is the probability they are all the same flavor?

(b) What is the probability they are all different flavors?

2.2.4■ Your Starburst candy has 12 pieces, three pieces of each of four flavors: berry, lemon, orange, and cherry, arranged in a random order in the pack. You draw the first four pieces from the pack.

(a) What is the probability $P[F_1]$ they are all the same flavor?

(b) What is the probability $P[F_4]$ they are all different flavors?

(c) What is the probability $P[F_2]$ that your Starburst has exactly two pieces of each of two different flavors?

2.2.5● In a game of rummy, you are dealt a seven-card hand.

(a) What is the probability $P[R_7]$ that your hand has only red cards?

(b) What is the probability $P[F]$ that your hand has only face cards?

(c) What is the probability $P[R_7F]$ that your hand has only red face cards? (The face cards are jack, queen, and king.)

2.2.6■ In a game of poker, you are dealt a five-card hand.

(a) What is the probability $P[R_5]$ that your hand has only red cards?

(b) What is the probability of a "full house" with three-of-a-kind and two-of-a-kind?

2.2.7● Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. How many different code words are there? How many code words have exactly three 0's?

2.2.8● Consider a language containing four letters: A, B, C, D. How many three-letter words can you form in this language? How many four-letter words can you form if each letter appears only once in each word?

2.2.9● On an American League baseball team with 15 field players and 10 pitchers, the manager selects a starting lineup with 8 field players, 1 pitcher, and 1 designated hitter. The lineup specifies the players for these positions and the positions in a batting order for the 8 field players and designated hitter. If the designated hitter must be chosen among all the field players, how many possible starting lineups are there?

2.2.10■ Suppose that in Problem 2.2.9, the designated hitter can be chosen from among all the players. How many possible starting lineups are there?

2.2.11● At a casino, the only game is numberless roulette. On a spin of the wheel, the ball lands in a space with color red (r), green (g), or black (b). The wheel has 19 red spaces, 19 green spaces and 2 black spaces.

(a) In 40 spins of the wheel, find the probability of the event

$$A = \{19 \text{ reds, } 19 \text{ greens, and } 2 \text{ blacks}\}.$$

(b) In 40 spins of the wheel, find the probability of $G_{19} = \{19 \text{ greens}\}$.

(c) The only bets allowed are red and green. Given that you randomly choose to bet red or green, what is the probability p that your bet is a winner?

2.2.12■ A basketball team has three pure centers, four pure forwards, four pure guards, and one swingman who can play either guard or forward. A pure position player can play only the designated position. If the coach must start a lineup with one center, two forwards, and two guards, how many possible lineups can the coach choose?

2.2.13♦ An instant lottery ticket consists of a collection of boxes covered with gray wax. For a subset of the boxes, the gray wax hides a special mark. If a player scratches off the correct number of the marked boxes (and no boxes without the mark), then that ticket is a winner. Design an instant lottery game in which a player scratches five boxes and the probability that a ticket is a winner is approximately 0.01.

2.3.1● Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. In each code word, a bit is a zero with probability 0.8, independent of any other bit.

(a) What is the probability of the code word 00111?

(b) What is the probability that a code word contains exactly three ones?

2.3.2● The Boston Celtics have won 16 NBA championships over approximately 50 years. Thus it may seem reasonable to assume that in a given year the Celtics win the title with probability $p = 16/50 = 0.32$, independent of any other year. Given such a model, what would be the probability of the Celtics winning eight straight championships beginning in 1959? Also, what would be the probability of the Celtics winning the title in 10 out of 11 years, starting in 1959? Given your answers, do you trust this simple probability model?

2.3.3● Suppose each day that you drive to work a traffic light that you encounter is either green with probability 7/16, red with probability 7/16, or yellow with probability 1/8, independent of the status of the light on any other day. If over the course of five days, G, Y, and R denote the number of times the light is found to be green, yellow, or red, respectively, what is the probability that $P[G = 2, Y = 1, R = 2]$? Also, what is the probability $P[G = R]$?

2.3.4■ In a game between two equal teams, the home team wins with probability $p > 1/2$. In a best of three playoff series, a team with the home advantage has a game at home, followed by a game away, followed by a home game if necessary. The series is over as soon as one team wins two games. What is $P[H]$, the probability that the team with the home advantage wins the series? Is the home advantage increased by playing a three-game series rather than a one-game playoff? That is, is it true that $P[H] \geq p$ for all $p \geq 1/2$?

2.3.5♦ A collection of field goal kickers are divided into groups 1 and 2. Group i has $3i$ kickers. On any kick, a kicker from group i will kick a field goal with probability $1/(i+1)$, independent of the outcome of any other kicks.

(a) A kicker is selected at random from among all the kickers and attempts one field goal. Let K be the event that a field goal is kicked. Find $P[K]$.

(b) Two kickers are selected at random; K_j is the event that kicker j kicks a field goal. Are K_1 and K_2 independent?

(c) A kicker is selected at random and attempts 10 field goals. Let M be the number of misses. Find $P[M = 5]$.

2.4.1 ■ A particular operation has six components. Each component has a failure probability q, independent of any other component. A successful operation requires both of the following conditions:

- Components 1, 2, and 3 all work, or component 4 works.
- Component 5 or component 6 works.

Draw a block diagram for this operation similar to those of Figure 2.2 on page 53. Derive a formula for the probability $P[W]$ that the operation is successful.

2.4.2 ■ We wish to modify the cellular telephone coding system in Example 2.21 in order to reduce the number of errors. In particular, if there are two or three zeroes in the received sequence of 5 bits, we will say that a deletion (event D) occurs. Otherwise, if at least 4 zeroes are received, the receiver decides a zero was sent, or if at least 4 ones are received, the receiver decides a one was sent. We say that an error occurs if i was sent and the receiver decides $j \neq i$ was sent. For this modified protocol, what is the probability $P[E]$ of an error? What is the probability $P[D]$ of a deletion?

2.4.3 ■ Suppose a 10-digit phone number is transmitted by a cellular phone using four binary symbols for each digit, using the model of binary symbol errors and deletions given in Problem 2.4.2. Let C denote the number of bits sent correctly, D the number of deletions, and E the number of errors. Find $P[C = c, D = d, E = e]$ for all c, d, and e.

2.4.4 ♦ Consider the device in Problem 2.4.1. Suppose we can replace any one component with an ultrareliable component that has a failure probability of $q/2 = 0.05$. Which component should we replace?

2.5.1 ■ Build a MATLAB simulation of 50 trials of the experiment of Example 2.3. Your output should be a pair of 50×1 vectors $\mathbf{C}$ and $\mathbf{H}$. For the ith trial, H_i will record whether it was heads ($H_i = 1$) or tails ($H_i = 0$), and $C_i \in \{1, 2\}$ will record which coin was picked.

2.5.2 ■ Following Quiz 2.3, suppose the communication link has different error probabilities for transmitting 0 and 1. When a 1 is sent, it is received as a 0 with probability 0.01. When a 0 is sent, it is received as a 1 with probability 0.03. Each bit in a packet is still equally likely to be a 0 or 1. Packets have been coded such that if five or fewer bits are received in error, then the packet can be decoded. Simulate the transmission of 100 packets, each containing 100 bits. Count the number of packets decoded correctly.

2.5.3 ■ For a failure probability $q = 0.2$, simulate 100 trials of the six-component test of Problem 2.4.1. How many devices were found to work? Perform 10 repetitions of the 100 trials. What do you learn from 10 repetitions of 100 trials compared to a simulated experiment with 100 trials?

2.5.4 ♦ Write a MATLAB function

```
N=countequal(G,T)
```

that duplicates the action of `hist(G,T)` in Example 2.26. Hint: Use `ndgrid`.

2.5.5 ♦ In this problem, we use a MATLAB simulation to "solve" Problem 2.4.4. Recall that a particular operation has six components. Each component has a failure probability q independent of any other component. The operation is successful if both

- Components 1, 2, and 3 all work, or component 4 works.
- Component 5 or component 6 works.

With $q = 0.2$, simulate the replacement of a component with an ultrareliable component. For each replacement of a regular component, perform 100 trials. Are 100 trials sufficient to decide which component should be replaced?